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Heel protector and corresponding rehabilitation systems and methods for using the same

Abstract

A device (**900**) includes a leg engaging section (**101**) and a foot engaging section (**102**) intersecting at a heel receiver (**103**). The leg engaging section and the foot engaging section defining a leg insertion aperture (**104**). A first compression wrap member (**301**) and a second compression wrap member (**302**) extend from the leg engaging section. An inflatable bladder (**501**) can be disposed along the leg engaging section between the leg insertion aperture and a compressible cushion layer (**902**). The inflatable bladder can be selectively inflatable through a connection tube (**502**) exiting the inflatable bladder at a non-orthogonal angle (**504**) relative to an edge (**507**) of the inflatable bladder. The leg engaging section can define at least one channel to permit the connection tube to exit the device.

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Background/Summary

CROSS REFERENCE TO PRIOR APPLICATIONS (1) This application is a continuation application from, and therefore claims priority to, U.S. Ser. No. 16/859,076, filed Apr. 27, 2020, which is a continuation application from, and therefore claims priority to U.S. Ser. No. 15/677,958, filed Aug. 15, 2017, which is a continuation application from, and therefore claims priority to, U.S. Ser. No. 14/206,395, filed Mar. 12, 2014, which claims priority and benefit under 35 U.S.C. § 119(e) from U.S. Provisional Application No. 61/781,682, filed Mar. 14, 2013, each of which is incorporated herein by reference. U.S. Ser. No. 14/206,395 is a continuation-in-part of, and therefore claims priority to, U.S. Ser. No. 13/757,233, filed Feb. 1, 2013, which is a continuation-in-part of, and therefore claims priority to, U.S. Ser. No. 13/649,920, filed Oct. 11, 2012, each of which is incorporated by reference for all purposes.

BACKGROUND

Technical Field

(1) This disclosure relates generally to therapy systems, and more particularly to devices for preventing complications during therapy.

Background Art

(2) Limb protection devices, including boots, braces, wraps, socks, sleeves, and the like are used to protect a patient's limbs. These devices can be used for a variety of reasons, including limb elevation, limb pressure alleviation, limb protection, and limb strengthening.

(3) While many of these devices work reasonably well in practice, problems with their usage exist. When left on for long periods of time, or when used incorrectly, these devices can sometimes lead to skin breakdown or the formation of pressure ulcers. Where this occurs, the therapeutic device creates new medical conditions that must be treated while aiding in the rehabilitation of previously existing conditions. These new issues only serve to extend the overall rehabilitation time for the patient. Accordingly, it would be advantageous to have an improved therapeutic device.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying figures, where like reference numerals refer to identical or functionally similar elements throughout the separate views and which together with the detailed description below are incorporated in and form part of the specification, serve to further illustrate various embodiments and to explain various principles and advantages all in accordance with the present disclosure.

(2) FIGS. 1 and 2 illustrate one explanatory portion of a device in accordance with one or more embodiments of the disclosure.

- (3) FIG. 3 illustrates another explanatory portion of a device in accordance with one or more embodiments of the disclosure.
- (4) FIG. 4 illustrates an explanatory device in accordance with one or more embodiments of the disclosure.
- (5) FIG. 5 illustrates another explanatory portion of a device in accordance with one or more embodiments of the disclosure.
- (6) FIG. 6 illustrates another explanatory portion of a device in accordance with one or more embodiments of the disclosure.
- (7) FIG. 7 illustrates another explanatory portion of a device in accordance with one or more embodiments of the disclosure.
- (8) FIG. 8 illustrates another explanatory portion of a device in accordance with one or more embodiments of the disclosure.
- (9) FIG. 9 illustrates one explanatory device in accordance with one or more embodiments of the disclosure.
- (10) FIG. 10 illustrates one explanatory device in accordance with one or more embodiments of the disclosure.
- (11) FIG. 11 illustrates one explanatory device in accordance with one or more embodiments of the disclosure.
- (12) FIG. 12 illustrates one explanatory device in accordance with one or more embodiments of the disclosure.
- (13) FIG. 13 illustrates one explanatory device in accordance with one or more embodiments of the disclosure.
- (14) FIG. 14 illustrates one explanatory device in accordance with one or more embodiments of the disclosure.
- (15) FIG. 15 illustrates one explanatory device in accordance with one or more embodiments of the disclosure.
- (16) FIG. 16 illustrates one explanatory device in accordance with one or more embodiments of the disclosure.
- (17) FIG. 17 illustrates one explanatory device in accordance with one or more embodiments of the disclosure.
- (18) FIG. 18 illustrates a patient's limb wearing one explanatory device in accordance with one or more embodiments of the disclosure.
- (19) FIG. 19 illustrates one explanatory device configured in accordance with one or more embodiments of the disclosure upon being applied to a patient's limb.
- (20) FIG. 20 illustrates one explanatory device in accordance with one or more embodiments of the disclosure upon being applied to a patient's limb.
- (21) Skilled artisans will appreciate that elements in the figures are illustrated for simplicity and clarity and have not necessarily been drawn to scale. For example, the dimensions of some of the elements in the figures may be exaggerated relative to other elements to help to improve understanding of embodiments of the present disclosure.

DETAILED DESCRIPTION OF THE DRAWINGS

(22) Embodiments of the disclosure are now described in detail. Referring to the drawings, like numbers indicate like parts throughout the views. As used in the description herein and throughout the claims, the following terms take the meanings explicitly associated herein, unless the context clearly dictates otherwise: the meaning of “a,” “an,” and “the” includes plural reference, the meaning of “in” includes “in” and “on.” Relational terms such as first and second, top and bottom, and the like may be used solely to distinguish one entity or action from another entity or action without necessarily requiring or implying any actual such relationship or order between such entities or actions. Also, reference designators shown herein in parenthesis indicate components shown in a figure other than the one in discussion. For example, talking about a device (10) while

discussing figure A would refer to an element, **10**, shown in figure other than figure A.

(23) Embodiments of the present disclosure provide a cushioned device that is configured both for compression therapy and for providing protection to a person's leg, foot, or heel during rehabilitation. In one embodiment, a device includes a leg engaging section and a foot engaging section. The leg engaging section and the foot engaging section intersect at a heel receiver. In one embodiment, the leg engaging section and the foot engaging section define a leg insertion aperture into which a patient's leg may be placed. To provide protection for the patient's leg during treatment or rehabilitation, in one embodiment the leg engaging section includes a compressible cushion layer. The compressible cushion layer can be an organic batting, such as a cotton or wool batting, or may alternatively be an inorganic batting, such as a poly fiber fill, compressible foam, or a synthetic material. Of course, combinations of organic batting materials and inorganic batting materials may also be used.

(24) To provide compression therapy concurrently with the delivery of protection, in one embodiment the device has integrated therewith a first compression wrap member and a second compression wrap. The first compression wrap member and the second compression wrap member each extend from the leg engaging section. In one embodiment the first compression wrap member and the second compression wrap member are disposed internal to the device, with side portions of the leg engaging section disposed outside the first compression wrap member and the second compression wrap member. In this configuration, the first compression wrap member and the second compression wrap member can be secured about the patient's limb prior to wrapping the soft, cushioned sides of the device around the limb.

(25) In one embodiment, an inflatable bladder is disposed along the leg engaging section. In one embodiment, the inflatable bladder is disposed between the leg insertion aperture and the compressible cushion layer so as to be adjacent to the patient's limb. Said differently, positioning the inflatable bladder between the patient's limb and the compressible cushion layer ensures compression occurs when the first compression wrap member and second compression wrap member are secured about the patient's leg.

(26) In one embodiment, the inflatable bladder is selectively inflatable through a connection tube. Embodiments of the disclosure contemplate that devices in accordance with embodiments of the disclosure can be worn by patients that are bed ridden for given amounts of time to reduce pressure ulcers from forming on the lower legs and feet. At the same time, embodiments provide the first compression wrap member and the second compression wrap member, which working with the inflatable bladder, provides compression therapy.

(27) Compression therapy may be required to prevent deep vein thrombosis (DVT) or venous thrombo-embolisms, which are conditions where clots form in the blood. Patients undergoing surgery, under anesthesia, or undergoing extended periods of bed rest are at risk of clotting conditions associated with DVT. The clotting conditions frequently occur in the deep veins of the lower extremities, such as in the lower legs, due to the tendency of blood to accumulate or pool in these areas. Static pools of blood can give rise to clotting conditions. Where clots form, circulation can be compromised, thereby putting the patient's health at risk. Further, clots can break free, which puts the patient at risk for embolism, which in some circumstances can be life threatening. Application of a compression device can work to prevent pooling, thereby reducing the risk that a clot will form.

(28) However, the inventors of embodiments of the present disclosure have come to understand that the use of a compression device with a conventional heel offloading boot or limb covering creates new problems. Specifically, when using compression devices with boots or other coverings there is an increased risk of skin breakdown due to the fact that tubing from the compression device can come into contact with the patient's skin when the boot or other covering is wrapped about the compression device. Prior art boots and coverings provide as much as three inches along which tubing from compression devices can contact the patient's skin. When the tubing is not placed

precisely within the boot or covering, it will contact the patient's skin, thereby significantly increasing the risk of skin breakdown. Even when the tubing is placed correctly when the boot or covering is applied, patient movement or tubing manipulation outside the boot or covering can cause the tubing to contact the patient's skin, thereby exacerbating skin breakdown. This problem can be exacerbated due to the fact that the boot or covering is applying pressure that presses the tubing against the patient's skin.

(29) Prior art compression therapy devices can also create unnecessary pressure points, thereby exacerbating the pressure ulcer occurrences. Moreover, traditional compression devices are themselves inherently stiff and apply large amounts of pressure to a patient's limb, which runs counter to the purpose of the pressure ulcers in a protective boot. The tubing of prior art systems is typically hard so as to not be crushed during use, and thereby serves as a prime source of pressure ulcers. As the patient moves around, the tubing can become compromised, can gather, and can bunch, thereby causing unwanted pressure on the skin.

(30) Embodiments of the disclosure provide a novel integrated device that is capable of providing pressure therapy while also providing working to prevent ulcers and other unnecessary maladies by providing cushioning and elevating features as well. In one embodiment, a device is configured as a cushioning boot that has at least a leg engaging section. The cushioning boot may have two sides and a foot engaging section. Each of the leg engaging section, the sides, and the foot engaging section can all include a compressible cushioning layer, manufactured from materials such as batting, foam, and the like, to reduce pressure on a patient's limb during extended bed rest. In one embodiment, the leg engaging section, and optionally the sides or the foot engaging section, also includes an inflatable bladder that can be coupled to an air pump to provide compression therapy.

(31) In one embodiment, the inflatable bladder includes an air port or air port portion that exits the inflatable bladder away from a patient's body and down portions of the device that prevent the port and/or tubing from contacting the patient's limb or skin. For example, in one embodiment the connection tube of the inflatable bladder exits the inflatable bladder at a non-orthogonal angle relative to an edge of the inflatable bladder. In one embodiment, the leg engaging section defines at least one channel to permit the connection tube to exit the device. This diverting path traversed by the port and/or tubing works to reduce the risk of the patient developing a pressure ulcer.

(32) In one embodiment, the inflatable bladder is configured under a top material layer of the leg engaging section. In one embodiment, the inflatable bladder may be disposed along an optional ancillary cushioning layer. Either the compressible cushion layer or the optional ancillary cushioning layer can provide support to reduce the risk of a patient developing pressure ulcers when the inflatable bladder is deflated. When the inflatable bladder is inflated, the cushion layer or the optional ancillary cushioning layer works to facilitate the proper blood circulation effect to the patient's limb, thereby functioning as a deep vein thrombosis (DVT) therapy sleeve. The cushion layer or the optional ancillary cushioning layer may be manufactured from foam. Alternatively, the cushion layer or the optional ancillary cushioning layer may be memory foam, general organic or inorganic batting, or organic or inorganic fill materials. In one embodiment, additional cushion layer or the optional ancillary cushioning layer material is disposed beneath the inflatable bladder. In one additional embodiment, a half inch or less of foam or other cushioning material can be disposed atop the bladder as well.

(33) In one embodiment, the leg engaging section intersects with the foot engaging section and a heel receiver. A leg insertion aperture is defined along the leg engaging section and a foot engaging section. Once the patient's limb is placed within the leg insertion aperture, the first compression wrap member and the second compression wrap member can be fastened about the patient's limb. The first compression wrap member and the second compression wrap member can be configured for providing compression therapy to a patient's limb. In one embodiment, the first compression wrap member and the second compression wrap member are manufactured from a material that can be elasticized and that has an outer face and an inner face. The inner face is disposed against the

patient's limb, while the outer face is visible when the wrap is applied to the limb. One of the first compression wrap member or the second compression wrap member defines a proximal edge, a distal edge, and first and second side edges. One of the side edges includes a plurality of attachment tabs that are configured to attach—by hook and loop fastener or other attachment device—to the outer face of the other of the first compression wrap member or the second compression wrap member when each is wrapped about the patient's limb.

(34) Where compression therapy is desired, the inflatable bladder can be inflated. For example, in one application the bladder can be inflated with air to a pressure of forty millimeters of mercury to apply pressure to a patient's limb for compression therapy. If compression therapy is not needed, the bladder can be left flaccid. Regardless of the state of the inflatable bladder, once the first compression wrap member and the second compression wrap member are secured, one or more fastening straps can wrap from one side of the leg engaging portion across the leg insertion aperture to another side of the leg engaging portion to retain the overall compound device on the patient's limb.

(35) In one or more embodiments, the device includes one or more apertures disposed along the leg engaging section. The apertures permit the connection tube extending from the inflatable bladder of a compression device to pass therethrough. Moreover, the connection tubing can be configured to exit the compression device at a non-orthogonal angle, thereby permitting the connection tubing to easily exit the device without risk of contacting the patient's skin.

(36) For example, in one embodiment, to provide a better user experience, the connection tube exits the inflatable bladder at a non-orthogonal angle relative to the distal edge of the inflatable bladder. When the inflatable bladder is disposed beneath the patient's leg, the non-orthogonal angle ensures that the connection tube does not run parallel to the patient's leg, thereby causing discomfort and potential skin breakdown that can occur if the connection tube passes along the patient's Achilles tendon. This angle can also facilitate the connection tube passing conveniently through the apertures in the medial or lateral sides of the leg engaging portion of the device. Advantageously, this both increases comfort for the patient over prior art designs and reduces or eliminates the risk of skin breakdown because the connection tube does not contact the patient's skin. Moreover, embodiments of the disclosure are easier for a health care services provider to apply.

(37) Turning now to FIGS. 1 and 2, illustrated therein is one explanatory portion **100** of a device configured in accordance with one or more embodiments of the disclosure. In the illustrative embodiment of FIGS. 1 and 2, the portion **100** includes a leg engaging section **101** and a foot engaging section **102**. The leg engaging section **101** intersects the foot engaging section **102** at a heel receiver **103**. In one embodiment, the heel receiver **103** defines an aperture **201** through which a patient's heel can be seen when the portion **100** is applied to the patient's leg. The leg engaging section **101** and the foot engaging section **102** have defined therealong a leg insertion aperture **104**. A patient's leg can be inserted through the leg insertion aperture **104**, as will be shown in FIG. 18 below.

(38) In one embodiment, the portion **100** includes one or more fastening straps **105,106,107,108** extending from the sides of the leg engaging section **101**, the foot engaging section, or combinations thereof. For example, in the illustrative embodiment of FIGS. 1 and 2, the portion **100** has four fastening straps **105,106,107,108** extending from its sides. At least one fastening strap **105** extends from a first side of the portion **100**, while others extend from another side of the portion **100**. This allows the fastening straps to “criss-cross” from one side of the portion **100** to the other. In this illustrative embodiment, two fastening straps **105,106** extend from the foot engaging section **102**, while two other fastening straps **107,108** extend from the leg engaging section **101**. Also, in this illustrative embodiment, three fastening straps **106,107,108** extend from the medial side **109** of the portion **100**, while one fastening strap **105** extends from the lateral side **110** of the portion **100**. This configuration is illustrative only, as other configurations and placements of the fastening straps **105,106,107,108** will be obvious to those of ordinary skill in the art having the

benefit of this disclosure.

(39) In one embodiment, the fastening straps **105,106,107,108** are stretchable. For example, they may comprise an elasticized material configured to stretch when being wrapped about the leg insertion aperture **104**. In another embodiment, the fastening straps **105,106,107,108** are not stretchable, but are rather material layers that are fixed in length and do not change when being wrapped about the leg insertion aperture **104**. The fastening straps **105,106,107,108** are affixed to the portion **100** by stitching in one embodiment. FIG. 2 illustrates fastening straps **107,108** being attached to the leg engaging section **101** along seam **202**.

(40) In one embodiment, each of the fastening straps **105,106,107,108** comprises one of a hook fastener or a loop fastener disposed therealong. Illustrating by example, fastening strap **105** may have hook fasteners disposed along side **111**. To complete the fastening system, in one embodiment the leg engaging section **101** includes one or more panels **112,113** that have a complementary fastener disposed therealong. Where, for example, fastening strap **105** includes hook fasteners, corresponding panel **112** may have loop fasteners disposed therealong, as the loop fasteners are complementary to the hook fasteners. Accordingly, when fastening strap **105** is wrapped across the leg insertion aperture **104**, it can be attached anywhere along panel **112**. The same is true with fastening straps **106,107,108** attaching to panel **113**. While hook and loop fasteners are one type of fastener or attachment mechanism suitable for use with embodiments of the disclosure, it should be noted that others will be obvious to those having ordinary skill in the art and the benefit of this disclosure. For example, the hook and loop fasteners can be replaced by laces, snaps, buttons, drawstrings, or other fastening devices.

(41) In one embodiment, the interior lining **114** of the central portion **121** of the leg engaging section **101** is soft and comfortable. For example, in one embodiment the interior lining **114** can be fleece or another soft material. In another embodiment, the interior lining **114** can be felt or chamois. As will be described below, in one or more embodiments the interior lining **114** can include an optional pocket (**1205**) into which an inflatable bladder may be inserted. In other embodiments the central portion **121** of the leg engaging section **101** includes an inflatable bladder disposed beneath the interior lining **114**.

(42) In one embodiment, the interior lining **114** has a relatively high coefficient of friction so that the portion **100** does not move when wrapped about a patient's limb or compression device attached thereto. For example, the interior lining **114** can be brushed, napped or sanded to raise its pile for comfort and increase the coefficient of friction. In one embodiment, the interior lining **114** has an antibacterial, antimicrobial, or anti-odor material integrated therein to help reduce the risk of bacteria, microbes, or odors from existing in the interior of the portion **100** after prolonged use. The interior lining **114** can also be manufactured from a wicking material. The exterior **116** of the portion **100** may be water resistant or waterproof as desired. In one embodiment, the interior of the portion **100** can be constructed from a cooling material, such as a gel that can be cooled to apply thermal therapy to the patient.

(43) As shown in FIG. 2, in one embodiment the leg engaging section **101** defines at least one aperture **203,204** disposed in an ankle region of the leg engaging section **101**. Various configurations of the aperture **203,204** will be described in more detail below with reference to FIGS. 13 and 14. As will also be described below, at least one aperture **203,204** advantageously allows connection tubes to exit the portion at non-orthogonal angles to reduce the possibility of pressure points arising under the patient's limb. Where a connection tube extends from the inflatable bladder, the inclusion of apertures **203,204** helps to minimize the risk of the connection tube contacting a patient's skin by providing an easy and convenient exit port. The addition of the apertures **203,204** on the lateral side **110** and medial side **109** of the leg engaging section **101** allows the connection tube emanating from the compression device to run directly out of the portion **100**, thereby eliminating the need to "tuck" tubing into the seam of the boot and away from the skin. In one embodiment, one aperture **203** is disposed about forty-five degrees around the leg

engaging section **101** from the other aperture **204**.

(44) In one embodiment the aperture **204** can be configured as a channel to permit a connection tube extending from an inflatable bladder of a compression device to pass therethrough. The channel can optionally be reinforced about its perimeter. The channel can be disposed in-line with a seam of the portion **100** or can be proximally located with the seam.

(45) Turning now to FIG. **3**, illustrated therein are additional portions **300** of one or more devices configured in accordance with one or more embodiments of the disclosure. The portions **300** shown in FIG. **3** are to attach to the portion (**100**) shown in FIG. **1** to create the overall device. While the portion (**100**) of FIG. **1** is to provide cushioning support for a patient's limb, the portions **300** of FIG. **3** can be included in the device to provide compression therapy.

(46) The portions of FIG. **3** include a first compression wrap member **301** and a second compression wrap member **302**. The first compression wrap member **301** and the second compression wrap member **302** are configured to attach to the portion (**100**) of FIG. **1** so as to wrap about the leg or other limb of a patient. While a leg is used as an explanatory limb for the purposes of discussion, those of ordinary skill in the art having the benefit of this disclosure will appreciate that the first compression wrap member **301** and the second compression wrap member **302** could equally be configured as an arm cuff, a knee sleeve, or sleeve for another body part.

(47) In one embodiment, the first compression wrap member **301** and the second compression wrap member **302** are manufactured from a non-stretchable material. In other embodiments, the first compression wrap member **301** and the second compression wrap member **302** are manufactured from a stretchable, elasticized material. The first compression wrap member **301** and the second compression wrap member **302** can comprise one or more layers of material that are stitched together. For example, in one embodiment, the first compression wrap member **301** and the second compression wrap member **302** each comprise at least two layers of material that are stitched together along a perimeter **303,304**. Panels, e.g., tab panel **305**, can also be defined by stitching **306** as well. The stitching **306** can be replaced by other suitable means for joining the materials, such as high frequency welds, ultrasonic welding, thermal bonding, heat-sealing, or adhesive bonding.

(48) One example of a suitable material for the first compression wrap member **301** and the second compression wrap member **302** is nylon tricot. Nylon tricot is manufactured by machines that use a warp-knit pattern to weave nylon fiber. The fibers are typically woven across the width of the material layer in a zigzag pattern. The nylon tricot can be 100% nylon fiber, or can alternatively be a blend of nylon and other fibers, including rayon or cotton. Nylon tricot works well as the first compression wrap member **301** and the second compression wrap member **302** because it does not snag or run easily. Moreover, it can be manufactured in a variety of colors. Nylon tricot can also be machine-washed.

(49) Other materials can be used as the first compression wrap member **301** and the second compression wrap member **302** as well. For instance, the first compression wrap member **301** and the second compression wrap member **302** can be manufactured from one or more sheets of plastic, neoprene, rubber, foam, felt, polymers, resins, and/or natural fabric materials. In some embodiments, only some layers of the first compression wrap member **301** and the second compression wrap member **302** can be configured to be stretchy and elastic. For instance, the outer face of the first compression wrap member **301** and the second compression wrap member **302** can be manufactured from a stretchy material, such as tricot stretch fabric, while an inner face of the first compression wrap member **301** and the second compression wrap member **302** is manufactured from a non-elastic material, or vice versa. Additionally, the various layers of first compression wrap member **301** and the second compression wrap member **302** may be manufactured from materials having varying degrees of elasticity or stretchiness.

(50) In the illustrative embodiment of FIG. **3**, the first compression wrap member **301** and the second compression wrap member **302** and each corresponding outer face define a proximal edge **307,308**, a distal edge **309,310**, a first side edge **312,313**, and a second side edge **314,315**. In this

embodiment, the second side edge **314** of the first compression wrap member **301** defines a plurality of attachment tabs **316,317,318**. In one embodiment, the attachment tabs **316,317,318** attach to the outer face of the second compression wrap member **302**. In one embodiment the attachment tabs **316,317,318** employ hook and loop fastening devices for attachment. For example, each of the attachment tabs **316,317,318** can include hook fasteners disposed on the inner face, while the outer face of the second compression wrap member **302** comprises loop pile fabric to which the hook fasteners can attach. It will be obvious to those of ordinary skill in the art having the benefit of this disclosure that other attachment mechanisms can be used, such as zippers, buttons, straps, laces, adhesive, or other devices.

(51) In one or more embodiments, the first side edge **312,313** and the second side edge **314,315** are not parallel. This is due to the fact that a medial reference line **320,321** extending across each of the first compression wrap member **301** and the second compression wrap member **302** has a curvature configured to facilitate the first compression wrap member **301** and the second compression wrap member **302** wrapping around a patient's limb. This curvature causes both the first side edge **312,313** and second side edge **314,315** to be oblique relative to each other so as to be substantially orthogonal with the medial reference line **320,321**. Accordingly, the longitudinal boundaries of the second side edges **314,315** form a quasi-frustoconical shape ("quasi" because the top and bottom are curved in accordance with the curvature).

(52) Turning now to FIG. **4**, illustrated therein is the first compression wrap member **301** and the second compression wrap member **302** coupled to the portion **100** of FIG. **1**. In this embodiment, the first compression wrap member **301** and the second compression wrap member **302** are coupled interior to the portion **100**. The first side (**312**) of the first compression wrap member is coupled to a seam **401** of the portion **100** between a first side **402** of the portion **100** and a central panel **403** of the portion **100**. Similarly, the first side (**303**) of the second compression wrap member **302** is coupled to a seam **404** of the portion **100** between a second side **405** of the portion and the central panel **403** of the portion **100**. This results in the first compression wrap member **301** and the second compression wrap member **302** being disposed interior to the first side **402** and the second side **405** of the leg engaging section **101** of the portion **100** in this illustrative embodiment. The first compression wrap member **301** and the second compression wrap member **302** are configured as compression straps extending from the central panel **403** interior to the first side **402** and the second side **405** in this illustrative embodiment.

(53) As shown, a hook fastener **406** is disposed along an interior side of the first compression wrap member **301**. A loop fastener **407** is disposed along an exterior side of the second compression wrap member **302**. The hook fastener **406** can couple to the loop fastener **407** when the first compression wrap member **301** and the second compression wrap member **302** are wrapped around a patient's limb.

(54) Turning now to FIGS. **5-8**, illustrated therein are different configurations for the central panel **403** of the portion (**100**) in accordance with one or more embodiments of the disclosure. Each view illustrates components of the assembly of FIG. **4** that are not generally visible from the assembly's exterior.

(55) Beginning with FIG. **5**, in one embodiment the central panel **403** of the portion (**100**) includes an inflatable bladder **501** that is configured to be selectively inflatable or deflatable. In one embodiment, the inflatable bladder **501** is disposed beneath the fabric of the central panel **403**. In another embodiment, the inflatable bladder **501** is disposed within a pocket (**1205**) of the central panel **403**. In the illustrative embodiment of FIG. **5**, the former is the case, as the inflatable bladder **501** is disposed between seam **401** and seam **404**. In one embodiment, seam **401** and seam **404** define the inflatable bladder **501**. In another embodiment, the inflatable bladder **501** is a separate component that is held in place between seam **401** and seam **404**. While the central panel **403** is one suitable location for the inflatable bladder **501**, it is illustrative only. Other locations will be obvious to those of ordinary skill in the art having the benefit of this disclosure. Disposing the

inflatable bladder **501** along the central panel **403** allows the inflatable bladder **501** to be positioned beneath the calf muscle of a patient who is lying upon their back.

(56) While the inflatable bladder **501** is shown illustratively in FIG. 5 as being a single chamber bladder with no internal welds or chambers, it should be understood that the inflatable bladder **501** may also be constructed as a multi-chamber bladder as well. Turning briefly to FIG. 7, illustrated therein is an alternate central panel **403** that includes multiple inflatable bladders **601,602,603**. A first inflatable bladder **601** is selectively inflatable through a connection tube **604** exiting the inflatable bladder **601** at a non-orthogonal angle, while an additional inflatable bladder **602** is also selectively inflatable through an additional connection tube **605** exiting the additional inflatable bladder **602** at another non-orthogonal angle.

(57) In one or more embodiments, the inflatable bladders **601,602,603** are selectively inflatable. Said differently, the first inflatable bladder **601** can be inflated at a first time while the additional inflatable bladder **602** can be inflated at another time, and so forth. It is contemplated that in some situations therapy may be improved by inflating the inflatable bladders **601,602,603** at different times. To provide this functionality, multiple connection tubes **604,605,606** extend from each inflatable bladder **601,602,603** for connection to a pump. In the illustrative embodiment of FIG. 7, each connection tube **604,605,606** extends down and away from the corresponding inflatable bladder **601,602,603**, i.e., at a non-orthogonal angle **607** relative to an edge **608** of the inflatable bladder **603**, so as to extend toward the foot engaging section (**102**) of the assembly of FIG. 4 for connection to a pump. A secondary function of this down and away orientation of the connection tubes **604,605,606** is stabilization, as this angular configuration helps to prevent devices in accordance with one or more embodiments from rolling.

(58) Turning now back to FIG. 5, in one embodiment the inflatable bladder **501** is selectively inflatable through a connection tube **502**. For example, in one application the inflatable bladder **501** can be inflated with air to a pressure of forty millimeters of mercury to apply pressure to a patient's limb for compression therapy. The connection tube **502** is coupled to the inflatable bladder **501** by way of a connector **503**.

(59) In one embodiment, to provide a more comfortable user experience, the connector **503** and connection tube **502** exit the inflatable bladder **501** at a non-orthogonal angle **504** relative to the edge **507** of the inflatable bladder **501**. For example, in one embodiment the non-orthogonal angle **504** is about 120 degrees. When the central panel **403** is disposed beneath the patient's leg, for instance, the non-orthogonal angle **504** ensures that the connection tube **402** does not run parallel to the patient's leg, thereby causing discomfort that occurs when the connection tube passes along the patient's Achilles tendon. The non-orthogonal angle **504** causes the connection tube **502** to naturally curve away from the patient's leg, thereby increasing the patient's comfort when using the devices in accordance with embodiments of the disclosure. While 120 degrees is one example of a suitable non-orthogonal angle, others will be obvious to those of ordinary skill in the art having the benefit of this disclosure. As shown in FIG. 6, the connection tube **502** can exit the inflatable bladder **501** at different locations so as to further increase the comfort of the patient.

(60) In one or more embodiments, the inflatable bladder **501** may be made of thin elastic material so that it expands and contracts with the introduction of cycling air. The shape of the inflatable bladder **501** may be square or trapezoidal. Alternatively, it may be configured as other patient limb-conforming shapes. The connector **503** may be an extension of the bladder material, which narrows as it extends from the main bladder portion in one or more embodiments. This configuration locates the connection tube **502**, which may be rigid, as far away from the patient's limb as possible.

(61) Turning now to FIG. 8, illustrated therein is another embodiment of the central panel **403**. As shown in FIG. 8, one embodiment of the central panel **403** comprises a foam layer **801**. The foam layer **801** serves as a cushion and may be disposed at different locations along the central panel **403**. In the illustrative embodiment of FIG. 8, the foam layer **801** is disposed adjacent to the bottom edge **802** of the central panel **403**. In one embodiment, the foam layer **801** extends distally from the

bottom edge **802** across only a portion of the central panel **403**. In one or more embodiments, the foam layer **801** can be configured to cover the connector (**503**) of the inflatable bladder (**501**) to slightly elevate the patient's heel when devices configured in accordance with embodiments of the disclosure are in use. This elevation helps to ensure that the connector (**503**) of the inflatable bladder (**501**) does not become a pressure point against the patient's leg.

(62) In one or more embodiments, the foam layer **801** and one or more inflatable bladders (**501,601,602,603**) can be used in combination. In one embodiment, the foam layer **801** is disposed between the one or more inflatable bladders (**501,601,602,603**) and the patient's limb. In another embodiment, the one or more inflatable bladders (**501,601,602,603**) are disposed between the patient's limb and the foam layer **801**. Thus, assemblies shown in FIGS. 5-7 and the assembly shown in FIG. 8 can be used in combination.

(63) Turning now to FIGS. 9 and 10, illustrated therein is a device **900** configured in accordance with one or more embodiments of the disclosure. The device **900** is shown in FIG. 9 sectional view so that internal and external components can be seen. The device **900** is shown in an open view in FIG. 10.

(64) In this illustrative embodiment, the device **900** includes a leg engaging section **101** and a foot engaging section **102** intersecting at a heel receiver. The leg engaging section **101** and the foot engaging section **102** define a leg insertion aperture **104**. At least the leg engaging section includes a compressible cushion layer **902**. The compressible cushion layer **902** can be manufactured from one of an organic batting or an inorganic batting, or alternatively of combinations thereof.

(65) The device **900** includes the first compression wrap member **301** and the second compression wrap member **302**. In this embodiment, the first compression wrap member **301** and the second compression wrap member **302** extend from the leg engaging section **101**. Here, the the first compression wrap member **301** and the second compression wrap member **302** are disposed interior of a first side **402** and a second side **405** of the leg engaging section **101**.

(66) The device **900** includes an inflatable bladder **501** disposed along the leg engaging section **101** between the leg insertion aperture **104** and the compressible cushion layer **902**. As noted above, in one or more embodiments the inflatable bladder **501** is selectively inflatable through a connection tube **502** exiting the inflatable bladder **501** at a non-orthogonal angle (**504**) relative to an edge (**507**) of the inflatable bladder **501**.

(67) As best seen in FIG. 10, in one embodiment the leg engaging section **101** defines at least one channel to permit the connection tube **502** to exit the device **900**. The port of the inflatable bladder **501** can be routed though the channel in the device **900** and angled away from a patient's limb so that it does not induce pressure on the limb. The channel works to immobilize the connection tube **502**, thereby keeping it from the interior region into which a patient's limb is placed. The channel also prevents the connection tube **502** from moving or wandering internally within the device. The non-orthogonal angle (**504**) of exit allows for the connection tube **502** to selectively pivot as a patient's limb rests in the device **900**. Moreover, it allows the connection tube **502** to articulate with limb movement. The channel works to ensure that the connection tube **502** maintains its position relative to a patient's limb even when the patient moves around in one or more embodiments.

(68) Turning now to FIG. 11, illustrated therein are potential attachment points **1101,1102,1103** for first compression wrap member **301** and the second compression wrap member **302**. In one embodiment, the first compression wrap member **301** can be attached to the leg engaging section **101** at a first attachment point **1101**. For example, the first compression wrap member **301** can be sewn or otherwise attached to the leg engaging section **101** at the first attachment point **1101**. Similarly, the second compression wrap member **302** can be sewn or otherwise attached to the leg engaging section **101** at attachment point **1103**.

(69) In one embodiment, the first compression wrap member **301** and the second compression wrap member **302** are attached to each other or are formed from a single piece of material that is sewn or otherwise attached to the leg engaging section at attachment point **1102**. Connecting the first

compression wrap member **301** and the second compression wrap member **302** only at attachment point **1102** allows the first compression wrap member **301** and the second compression wrap member **302** to more tightly wrap about a patient's limb. Further, a seam disposed at attachment point **1102** does not present a pressure ulcer pressure point as it is covered by the inflatable bladder **501** and optionally additional padding **1104**.

(70) In one or more embodiments, the first compression wrap member **301** and the second compression wrap member **302** can be attached to the leg engaging section **101** at combinations of the attachment points **1101,1102,1103**. Additionally, other attachment points will be obvious to those of ordinary skill in the art having the benefit of this disclosure.

(71) Turning now to FIG. **12**, illustrated therein is an alternate device **1200** configured in accordance with one or more embodiments of the disclosure. To provide additional lateral stability, the device **1200** of FIG. **12** includes two foam or air-filled tubes **1201,1202** disposed within the leg engaging section **101**. The compressible cushion layer **902** surrounds the first tube **1201** and the second tube **1202** in this embodiment. As noted above, the compressible cushion layer **902** can comprise one of an organic batting, an inorganic batting, or combinations thereof.

(72) In this illustrative embodiment, the first tube **1201** disposed to a first side of a medial line **1203** of the leg engaging section **101**, while the second tube **1202** is disposed to a second side of the medial line **1203**. When a patient's limb is inserted into the leg insertion aperture **104**, placement of the limb on the leg engaging section **101** causes the first tube **1201** and the second tube **1202** to spread to either side of the patient's limb, thereby increasing stability.

(73) Turning now to FIGS. **13-16**, illustrated therein are different connection tubing configurations in accordance with embodiments of the disclosure. Beginning with FIG. **13**, illustrated therein is an exterior view of the device **900** of FIG. **9**. As shown, the device **900** includes a port **1301** extending from the channel of the device **900**. In one embodiment, the port **1301** extends from the channel at an angle **1302** relative to a side **1303** of the device **900**. The angle **1302** can match the angle (**504**) at which the connection tube (**502**) exits the inflatable bladder (**501**) in one or more embodiments.

(74) As shown in FIG. **14**, in one embodiment, a strain relief fitting **1401** can be disposed about the connection tube **1402** at an exterior of the device **900**. In one embodiment the strain relief fitting **1401** is disposed on the outside surface of the device to prevent the connection tube **1402** from sliding into channel, thereby forcing the connection tube (**502**) exiting the inflatable bladder (**501**) from translating into the interior of the leg insertion aperture. The strain relief fitting **1401** may also further direct the connection tube **1402**, which may simply be an extension of connection tube (**502**), towards the back of the device or at least towards the position of the pump which is typically at the foot of the bed in a hospital environment. The strain relief fitting **1401** may incorporate or be coupled to a right angle elbow connector (shown as element (**1204**) in FIG. **12**) that further connect to tubing that extends to the pump.

(75) Turning now to FIG. **15**, this device **1500** includes multiple inflatable bladders. Accordingly, multiple connection tubes **1501,1502,1503** exit the device **1500** due to the fact that the leg engaging section **1505** defines at least one additional channel to permit the at least one additional connection tube **1502,1503** to exit the device **1500**. In this illustrative embodiment, the connection tubes **1501,1502,1503** are coupled to a common tube **1504** exterior to the device **1500**. The common tube **1504** can then be connected to a pump. By contrast, in FIGS. **16-17**, the connection tubes **1601,1602,1603** are coupled to a common tube **1604** interior to the device **1600**. Tubes **1201,1202** are included to ensure that this interior connection does not cause pressure on the patient's leg.

(76) Turning now to FIG. **18**, illustrated therein is a method of using one or more devices configured in accordance with embodiments of the disclosure. When device **1800** is donned, the leg **1801** is laid into the receiving cavity **1802**. The compression straps **1803,1804** formed by the first compression wrap member (**301**) and the second compression wrap member (**302**) are pulled up and over the leg **1801** from either side. The compression straps **1803,1804** are then connected

together as described above. In one embodiment, the compression straps **1803,1804** are all that is needed to secure the device **1800** to the patient's leg **1801**. Use of the fastening straps **1805,1806** further secure the device **1800** to the leg **1801**.

(77) Where used, a health care services provider **1807** can then wrap the fastening straps **1805,1806** across the leg insertion aperture **1808** to retain the device to the patient's leg **1801**. The result of this wrapping is shown in FIG. **19**. As shown in FIG. **19**, the connection tube **1902** passes through aperture **1903**, thereby eliminating any opportunity for the connection tube **1902** to touch the patient's skin. This reduces the chances of skin breakdown while the patient is wearing the device **1800**.

(78) Accessories can be provided for devices in accordance with embodiments of the disclosure. Turning now to FIG. **20**, illustrated therein is one such example. A rehabilitation system includes a device **1800** in accordance with embodiments of the disclosure and a bolster **2001**. In this embodiment, the bolster **2001** has been placed beside the device **1800** to provide resistance to rotational motion of the patient's leg. Said differently, the bolster **2001** is configured to stabilize the device **1800** rotationally when worn by a patient.

(79) In this illustrative embodiment, the bolster **2001** is generally triangular in cross section and provides an “ambidextrous” stabilizing wedge that can be placed on either side of the device **1800**. In one embodiment, a health care services provider (**1807**) is instructed to place a first bolster on one side of the device **1800** and a second bolster on the other side of the device **1800**. In other embodiments, a single bolster **2001** can be used as shown in FIG. **20**. While a triangular cross section of the bolster **2001** is shown in this illustrative embodiment, other cross sectional shapes will be obvious to those of ordinary skill in the art having the benefit of this disclosure.

(80) In one embodiment, the bolster **2001** is attached to the device **1800**. For example an edge of the bolster **2001** can be stitched to a seam of the leg engaging section of the device **1800**. However, in other embodiments, the bolster **2001** can be completely separated from the device **1800** so as to be used only when circumstances warrant. In the illustrative embodiment of FIG. **20**, the bolster **2001** includes a fastener that is complementary to a fastener disposed on an exterior of the leg engaging section of the device **1800**. For example, where the leg engaging section includes one of a hook fastener or a loop fastener, a complementary fastener can be disposed on the exterior of the bolster **2001** to attach the two components together. In such a configuration, the bolster **2001** can be attached to the device **1800** as necessary, but can be removed when not needed.

(81) In this illustrative embodiment, the bolster **2001** has been configured with a channel **2002** configured to permit the connection tube **2003** to pass from the device **1800** through the channel **2002**. Accordingly, in this illustrative embodiment, the channel **2002** is configured with a shape that is complementary to that of the connection tube **2003**. Those of ordinary skill in the art having the benefit of this disclosure will realize that the channel **2002** could take any of a variety of shapes. For example, the channel **2002** may be much wider than the connection tube **2003** so as to permit the connection tube **2003** to be placed at various lateral locations without moving the bolster **2001**.

(82) In the foregoing specification, specific embodiments of the present disclosure have been described. However, one of ordinary skill in the art appreciates that various modifications and changes can be made without departing from the scope of the present disclosure as set forth in the claims below. Thus, while preferred embodiments of the disclosure have been illustrated and described, it is clear that the disclosure is not so limited. Numerous modifications, changes, variations, substitutions, and equivalents will occur to those skilled in the art without departing from the spirit and scope of the present disclosure as defined by the following claims. Accordingly, the specification and figures are to be regarded in an illustrative rather than a restrictive sense, and all such modifications are intended to be included within the scope of present disclosure. The benefits, advantages, solutions to problems, and any element(s) that may cause any benefit,

advantage, or solution to occur or become more pronounced are not to be construed as a critical, required, or essential features or elements of any or all the claims.

Claims

1. A device, comprising: a leg engaging section intersecting a foot engaging section, at least the leg engaging section comprising a compressible cushion layer; a first compression wrap member and a second compression wrap member, each attached to the leg engaging section at different locations interior to a first side of the leg engaging section and a second side of the leg engaging section; and one or more fastening straps, affixed to and extending from the leg engaging section; and the leg engaging section defining at least one channel to permit a connection tube to exit the device.
 2. The device of claim 1, further comprising one or more other fastening straps, affixed to, and extending from, the leg engaging section, the foot engaging section, or combinations thereof.
 3. The device of claim 2, the one or more fastening straps comprising at least one fastening strap extending from a first side of the device and at least one other fastening strap extending from another side of the device.
 4. The device of claim 2, the first compression wrap member comprising an edge defining a plurality of attachment tabs.
 5. The device of claim 1, further comprising an inflatable bladder that is selectively inflatable through a connection tube exiting the inflatable bladder at a non-orthogonal angle.
 6. The device of claim 5, the leg engaging section defining at least one additional channel to permit at least one additional connection tube to exit the device.
 7. The device of claim 6, further comprising at least one additional inflatable bladder, wherein the inflatable bladder and the at least one additional inflatable bladder are sequentially inflatable.
 8. The device of claim 6, the connection tube and the at least one additional connection tube coupled to a common tube exterior to the device.
 9. The device of claim 6, the connection tube and the at least one additional connection tube coupled to a common tube interior to the device.
 10. The device of claim 1, the compressible cushion layer comprising an organic batting.
 11. The device of claim 1, the compressible cushion layer comprising an inorganic batting.
 12. The device of claim 1, further comprising an inflatable bladder disposed within a sleeve of the leg engaging section.
 13. The device of claim 1, further comprising a port extending from the at least one channel.
 14. The device of claim 1, wherein the at least one channel is wider than the connection tube.
 15. The device of claim 1, further comprising a bolster.
 16. The device of claim 15, wherein the bolster is configured to stabilize the device rotationally.
 17. The device of claim 16, the device comprising a fastener disposed on an exterior of the leg engaging section and the bolster comprising a complementary fastener disposed on an exterior of the bolster.
 18. The device of claim 1, the one or more fastening straps comprising four fastening straps.
 19. The device of claim 1, the first compression wrap member and the second compression wrap member manufactured from a stretchable material.
 20. The device of claim 1, wherein the at least one channel is reinforced about its perimeter.
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